

MOBILE PRICE PREDICTION USING MACHINE LEARNING AND FLASK

A Mini Project Report
submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Engineering

in

Artificial Intelligence and Data Science

By

Pasunuri Kathyayini (1601-21-771-088)

Ambekar Tejas (1601-21-771-095)

Under the esteemed guidance of

Mr. P. Vasanth Sena

Assistant Professor



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
HYDERABAD – 500075**

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3. Possess human values, professional ethics, application-oriented skills, and engage in lifelong learning.
4. Contribute to the research community to meet the needs of public and private sectors.

PROGRAM SPECIFIC OUTCOMES (PSOs)

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2. Develop professional skills in the thrust areas like ANN and Deep learning, Robotics, Internet of Things and Big Data Analytics.
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1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems
2. **Problem analysis:** Identify, formulate, review, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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MINI PROJECT-II

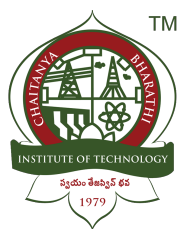
COURSE OBJECTIVES

1. To enable students learning by doing.
2. To develop capability to analyse and solve real world problems.
3. To inculcate innovative ideas of the students.
4. To impart team building and management skills among students.
5. To instill writing and presentation skills for completing the project.

COURSE OUTCOMES

Upon successful completion of this course, students will be able to:

1. Interpret Literature with the purpose of formulating a project proposal.
2. Plan, Analyse, Design and Implement a project using SDLC model.
3. Find the solution of identified problem with the help of modern Technology and give priority to real time scenarios.
4. Plan to work as a team and to focus on getting a working project done and submit a report within astipulated period of time.
5. Prepare and submit the Report and deliver presentation before the Departmental Committee.



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CO-PO MAPPING

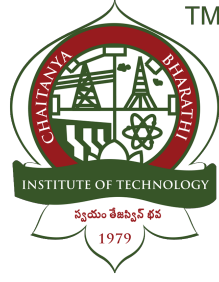
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CO1	3	3	2	3	3	3	3	2	1	2	3	3
CO2	3	3	3	3	3	3	3	2	1	2	3	3
CO3	3	3	3	3	3	3	3	2	-	2	3	3
CO4	2	2	2	3	3	3	3	2	3	3	2	3
CO5	1	2	1	2	3	3	-	-	2	3	-	-

Mapping of Course Outcomes with Program Outcomes

CO-PSO MAPPING

	PSO1	PSO2	PSO3
CO1	2	3	3
CO2	3	3	3
CO3	3	3	3
CO4	2	3	3
CO5	-	3	-

Mapping of Course Outcomes with Program Specific Outcomes



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DECLARATION CERTIFICATE

We hereby declare that the project titled **MOBILE PRICE PREDICTION USING MACHINE LEARNING AND FLASK** submitted by us to the **Artificial Intelligence and Data Science CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** in partial fulfillment of the requirements for the award of **Bachelor of Engineering** is a bona-fide record of the work carried out by us under the supervision of **Mr. P. Vasanth Sena** . We further declare that the work reported in this project, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

Project Associates

Pasunuri Kathyayini (1601-21-771-088)

Ambekar Tejas (1601-21-771-095)



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HYDERABAD – 500075**

BONAFIDE CERTIFICATE

This is to certify that the project titled **MOBILE PRICE PREDICTION USING MACHINE LEARNING AND FLASK** is a bonafide record of the work done by

Pasunuri Kathyayini (1601-21-771-088)

Ambekar Tejas (1601-21-771-095)

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Artificial Intelligence and Data Science** to the **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** carried out under my guidance and supervision during the year 2022-23. The results presented in this project report have not been submitted to any other university or Institute for the award of any degree.

Mr. P. Vasanth Sena

Guide

Dr.K.Ramana

Head of the Department

Submitted for Semester Mini-Project viva-voce examination held on _____

Examiner-1

Examiner-2

ABSTRACT

Mobile Price Prediction is an important need for people and companies who manufacture and sell mobile phones as it is a challenging task for them to determine the correct price of their mobile phone considering the variety of complex features, the specifications and the model. Accurately pricing their mobile phone before releasing it out for sale is critical because it determines their profitability and the market share. Pricing a mobile phone on par with its competitors and convincing people that it is worth the services their mobile offers is essential to establish a strong position in the highly competitive industry. The existing system for predicting mobile phone prices relies on human experts to analyze the features and specifications, which is time-consuming and prone to errors, leading to a great chance of inaccurate price predictions.

To address this challenge, this Mini Project aims to develop a Mobile Price Prediction system using Classification Algorithms that can help in predicting the mobile prices accurately. The proposed system leverages the power of Machine Learning Algorithms to analyze the data and make accurate predictions. This system increases the efficiency of the process, reduces the time and effort required in the existing method. It also eliminates the subjectivity and errors associated with the existing system, providing more reliable price predictions. The system made by us improves the competitiveness of the mobile sellers by enabling them to price the new mobile models they release accurately and strengthen their position in the market.

In order to implement this project, various tools and technologies are required, including programming languages such as Python, Data Analysis and Visualization libraries such as NumPy, Pandas, Matplotlib, and Machine Learning libraries like scikit-learn. Additionally, the system requires a user interface that allows sellers to input the features and specifications of the mobile phone, then view the predicted prices. This can be developed using HTML, CSS, and other web development tools such as Python web framework Flask, Python WSGI HTTP server Gunicorn.

Keywords : Prediction, Classification Algorithms, Machine Learning, Python, Data Analysis and Visualisation, Flask, HTML, CSS.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to the following people for guiding us through this course and without whom this project and the results achieved from it would not have reached completion.

Mr. P. Vasanth Sena, Assistant Professor, Department of Artificial Intelligence and Data Science, for helping us and guiding us in the course of this project. Without his guidance, we would not have been able to successfully complete this project. His patience and genial attitude is and always will be a source of inspiration to us.

Dr.K.Ramana, the Head of the Department, Department of Artificial Intelligence and Data Science, for allowing us to avail the facilities at the department.

We are also thankful to the faculty and staff members of the Department of Artificial Intelligence and Data Science, our individual parents and our friends for their constant support and help.

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CHAPTER 1

INTRODUCTION

In today's fast-paced world, mobiles have become an integral part of our lives. With numerous options available, consumers often face the challenge of selecting the right mobile device within their budget. It is also critical for the mobile manufacturers to put a reasonable and affordable price for the respective features as it directly impacts the profit of the manufacturer. Machine Learning algorithms offer a solution by predicting mobile prices based on various features and specifications. We employ these Machine Learning Algorithms to accurately predict prices of the mobile and in turn increase the profits and reliability of mobile manufacturing companies.

1.1 OVERVIEW

1. Aim of the project: The project aims to predict the price of a mobile phone based on its specifications using Machine Learning. Four different machine learning models, namely k-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, and Random Forest, are used for prediction. The dataset used for training the models contains 20 different mobile specifications and their corresponding price ranges.

2. Data Collection and Preprocessing: The first step involves collecting relevant data on various mobile phones, including features such as processor speed, RAM, internal storage, camera quality, battery capacity, etc. The dataset used in the project is sourced from Kaggle. The data is then preprocessed to handle missing values, normalize numerical features, and encode categorical variables if necessary. Valuable insights from the data have also been extracted to properly understand the relationships between different features of the dataset.

3. Model Training and Evaluation: The preprocessed data is divided into a training set and a testing set. The four machine learning models (KNN, Logistic Regression, Decision Tree, and Random Forest) are trained using the training data. The accuracy of each model is evaluated using metrics such as Accuracy score.

4. Model Selection: After training and evaluation, the model with the highest accuracy on the testing data is selected as the final model for mobile price prediction. In this case, KNN is chosen due to its high accuracy compared to the other models.

5. Flask Web Application: A Flask web application is developed to make the mobile price prediction accessible through a user-friendly interface. Flask is a lightweight web framework in Python. The application allows users to input the

specifications of a mobile phone they are interested in, and it will provide the predicted price as an output.

6. User Interaction: Users can access the Flask web app through their web browsers. They will be presented with an input form where they can fill in the specifications of the mobile phone they want to inquire about. Once the user submits the input, the Flask app will use the trained KNN model to predict the price based on the provided specifications.

7. Price Prediction and Output: The Flask app passes the user's input to the KNN model, which then predicts the price of the mobile phone. The predicted price is displayed as an output on the website for the user to view.

8. Continuous Improvement: The project can be continuously improved by collecting more data, experimenting with different machine learning algorithms, or fine-tuning hyperparameters of the models to achieve even better accuracy in mobile price prediction.

1.2 APPLICATIONS OF THE MODELS USED

KNN (K-Nearest Neighbors) is a simple and effective machine learning algorithm used for both classification and regression tasks. Its main principle is to find the "k" nearest data points to a given input sample and make predictions based on the majority class (for classification) or the mean/median of the nearest neighbors' target values (for regression). KNN finds numerous applications in various fields due to its simplicity and versatility. Some of the common applications of the KNN model include:

1. Image recognition: KNN can be used to classify images into different categories based on the similarity of their features with labeled training data. For example, it can classify handwritten digits or recognize different objects in images.

2. Anomaly detection: KNN can be employed to identify anomalies in data, such as fraudulent transactions in finance or defective products in manufacturing, by identifying data points that deviate significantly from the majority.

3. Recommender systems: KNN can be used to build user-based or item-based collaborative filtering systems for recommendation purposes, where similar users or items are used to suggest relevant products, movies, or content to users.

4. Medical diagnosis: KNN can assist in medical diagnosis by predicting the disease or condition of a patient based on the similarity of their symptoms and medical history with existing patient records.

5. Natural Language Processing (NLP): KNN can be used for text classification tasks such as sentiment analysis, spam detection, or topic categorization by measuring the similarity of text samples.

6. Customer segmentation: KNN can be applied to segment customers into different groups based on their purchasing behavior or demographic similarities, which can be used for targeted marketing campaigns.

7. Environmental monitoring: KNN can be used to classify environmental data, such as air quality, water quality, or species classification based on measurements from sensors and historical data.

8. Credit risk assessment: KNN can help assess the credit risk of loan applicants by comparing their financial information and credit history with historical loan data.

9. Collaborative filtering for social networks: KNN can be applied to recommend friends or connections in social networks by finding users with similar interests or interactions.

10. Speech recognition: KNN can be used in speech recognition systems to classify spoken words or phonemes based on their similarity with reference samples.

1.3 PROBELM STATEMENT

To create a Machine Learning model that is efficient and fast in predicting the price of the mobile accurately and building an interface for users to input the features and obtain the required output(price).

1.4 ORGANIZATION OF THE PROJECT

1.4.1 Project Planning and Requirements Gathering:

This module will define the project scope and objectives, gather requirements for the user interface module and the machine learning module and determines the mobile phone specifications to be included in the prediction model and the desired features of the user interface.

1.4.2 Data Collection and Preprocessing Module:

This module will gather data on mobile phone specifications and prices from various sources and perform data preprocessing tasks like handling missing values, normalization, and encoding categorical variables.

1.4.3 Machine Learning Module:

This module will include the implementation of four different models: k-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, and Random Forest. Each model will be trained using the preprocessed data and evaluated to determine the best-performing model (KNN).

1.4.4 Model Selection and Integration Module:

This module will be responsible for selecting the KNN model and integrating it into the Flask web application. It will handle the logic to choose the best-performing model based on evaluation results and connect it to the user interface module for predictions.

1.4.5 User Interface Module:

This module is to handle the user interface using Flask. Here, we will define routes, views, and templates to create a user-friendly interface for mobile price prediction. It will include input forms where users can enter mobile phone specifications.

1.4.6 Prediction Module:

Then, we create a module responsible for making predictions using the selected KNN model. This module will take input from the user interface, pass it to the KNN model, and return the predicted price.

1.4.7 Flask App Development and Integration:

Integrate the user interface, prediction module, and machine learning module into a Flask web application. Connect the prediction module to the machine learning module to make real-time price predictions based on user input.

1.4.8 Testing and Debugging:

Thoroughly test the user interface module, machine learning module, and their integration within the Flask app. Identify and fix any bugs or issues to ensure the application functions correctly.

By organizing the project into different modules, each responsible for specific functionalities, the development process has been more manageable, efficient, and maintainable. The clear separation of concerns also allowed for better collaboration among the team members and easier scalability for future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Paper-1: Prediction of Mobile Phone Price Class using Supervised Machine Learning Techniques by A Varun Kiran and Dr. Jebakumar R. SRM Institute of Science and Technology, Kattankulathur Campus, Chennai – 603203, Tamil Nadu, India.[1]

Abstract:- The aim of this research is to develop a model to predict the price of a mobile when the specifications of a mobile are given and to find the ML algorithm that predicts the price most accurately. The usage of archival data to accurately forecast forthcoming instances is the essence of Predictive Analytics. One of the ways Predictive Analytics can be performed is by using Machine Learning. Predictive Machine Learning works by taking in data as input to develop and train a prediction model and the trained model is used to predict the outcome of future data instances. Supervised Machine Learning algorithms make use of data that contains a pre-defined class label, which is the attribute that needs to be predicted. The class label is the price of a mobile in our case. The Mobile Price Class dataset sourced from the Kaggle data science community website (<https://www.kaggle.com/iabhishekofficial/mobileprice-classification>) that categorizes mobiles into price ranges was used to train the prediction model. Python is used due to its readily accessible ML libraries. Various classification algorithms were used to train the model to try and find the algorithm that is able to predict the mobile price class most accurately. Metrics like accuracy score, confusion matrix, etc. are used to evaluate the trained model to determine the algorithm most suitable among the ones used.

2.2 Paper-2: Mobile Price Class prediction using Machine Learning Techniques by Muhammad Asim, UET Lahore, Pakistan and Zafar Khan, UET Lahore, Pakistan[2]

Abstract:- To predict “If the mobile with given features will be Economical or Expensive” is the main motive of this research work. Real Dataset is collected from website www.GSMArena.com . Different feature selection algorithms are used

to identify and remove less important and redundant features and have minimum computational complexity. Different classifiers are used to achieve as higher accuracy as possible. Results are compared in terms of highest accuracy achieved and minimum features selected. Conclusion is made on the base of best feature selection algorithm and best classifier for the given dataset. This work can be used in any type of marketing and business to find optimal product(with minimum cost and maximum features). Future work is suggested to extend this research and find more sophisticated solution to the given problem and more accurate tool for price estimation.

2.3 PREREQUISITES

2.3.1 BASICS OF MODELS

The below is a list of some basic Machine Learning Models:

1. k-Nearest Neighbors (KNN): K-Nearest Neighbors is a simple and intuitive algorithm used for classification and regression tasks. In KNN, the prediction is made based on the majority class of the k-nearest data points to the input instance in the feature space. It's a non-parametric algorithm, meaning it doesn't make assumptions about the underlying data distribution.

2. Logistic Regression: Logistic Regression is a widely used algorithm for binary classification problems. It models the probability that an instance belongs to a particular class using a logistic function. The algorithm learns a linear relationship between the features and the log-odds of the target class, making it a linear model for binary classification tasks.

3. Decision Tree: A Decision Tree is a hierarchical structure that recursively partitions the data based on the values of the features to make predictions. Each internal node represents a decision based on a specific feature, and each leaf node represents the predicted class or value. Decision Trees are popular due to their interpretability and ability to handle both classification and regression tasks.

4. Random Forest: Random Forest is an ensemble learning method that combines multiple decision trees to improve accuracy and reduce overfitting. It creates a set of decision trees during training and combines their predictions through voting (for classification) or averaging (for regression). Random Forest is robust, handles high-dimensional data well, and is less prone to overfitting compared to individual decision trees.

5. Support Vector Machines (SVM): SVM is a powerful algorithm for both classification and regression tasks. It finds the hyperplane that best separates the data into different classes while maximizing the margin between the classes. SVM

is effective in high-dimensional spaces and can handle complex decision boundaries through the use of kernel functions.

6. Neural Networks: Neural Networks are a class of deep learning models inspired by the structure and functioning of the human brain. They consist of multiple interconnected layers of artificial neurons (nodes) that learn to extract features from the data and make predictions. Neural Networks have shown remarkable success in various tasks like image recognition, natural language processing, and more.

These are just a few of the fundamental ML models used in various applications. Each model has its strengths and weaknesses, and the choice of the model depends on the nature of the problem, the available data, and the desired accuracy and interpretability of the results.

2.3.2 BASICS OF HTML

HTML (Hypertext Markup Language) is the fundamental building block of web development. It is a standard markup language used to create the structure and content of web pages. HTML utilizes a system of tags and elements to define various components of a webpage, such as headings, paragraphs, images, links, and more. Each HTML element is represented by an opening tag (`< tag >`) and a closing tag (`< /tag >`), and they work together to enclose and define the content within.

An HTML document typically begins with a `<!DOCTYPE >` declaration, followed by the `< html >` element, which serves as the root element of the page. The content is further organized into sections, with the `< head >` section containing metadata like the page title, character set, and external resources, while the `< body >` section holds the visible content displayed to users. HTML allows web developers to create structured, semantically meaningful content, enabling web browsers to interpret and render web pages appropriately. When combined with CSS (Cascading Style Sheets) and JavaScript, HTML forms the backbone of modern web development, providing the basis for designing visually appealing, interactive, and responsive websites across various platforms and devices.

2.3.3 BASICS OF CSS

CSS (Cascading Style Sheets) is a crucial component of web development, working in tandem with HTML to define the visual presentation and layout of web pages. By utilizing CSS, web designers can apply styles and formatting to HTML elements, such as colors, fonts, margins, and spacing. CSS allows for the separation of content from presentation, making it easier to maintain and update the design of a website without altering its underlying structure.

CSS operates through a set of rules, where selectors target HTML elements, and declarations define the styles to be applied. Styles can be specified within an HTML document using the `< style >` tag, or in an external CSS file linked to the HTML page. With CSS, designers have the flexibility to create responsive layouts that adapt to different screen sizes and devices, ensuring a consistent and visually appealing user experience across desktops, tablets, and mobiles. Additionally, CSS3, the latest version of CSS, introduces advanced features like animations, transitions, and media queries, further enhancing the creative possibilities for designing modern and interactive web interfaces.

2.3.4 BASICS OF FLASK

Flask is a lightweight and flexible web framework written in Python. It is designed to make web development simple and efficient, allowing developers to build web applications quickly and with minimal boilerplate code. Flask follows the WSGI (Web Server Gateway Interface) standard, making it compatible with various web servers and platforms.

With Flask, developers can create web applications by defining routes and views that handle incoming requests and generate responses. It supports various HTTP methods like GET, POST, PUT, DELETE, etc., allowing the creation of RESTful APIs. Flask also offers template rendering capabilities, enabling the dynamic generation of HTML pages with embedded Python code. Additionally, Flask supports extensions that provide additional functionalities such as database integration, form handling, and user authentication. Due to its simplicity, flexibility, and extensive documentation, Flask is a popular choice for web developers, especially for small to medium-sized projects, prototypes, and APIs.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 FUNCTIONAL REQUIREMENTS

1. User Input: The web application should provide an input form where users can enter mobile phone specifications, such as processor speed, RAM, camera quality, storage capacity, and other relevant features.

2. Real-time Prediction: The application should use the trained KNN model to provide real-time predictions of mobile phone prices based on the user's input.

3. Model Selection: The application should automatically select the KNN model as the default machine learning algorithm for price prediction due to its higher accuracy.

4. User Interface: The user interface should be intuitive, user-friendly, and responsive to accommodate users across different devices and screen sizes.

5. Error Handling: The application should handle errors gracefully, displaying clear error messages if the user enters incorrect or incomplete specifications.

3.2 NON-FUNCTIONAL REQUIREMENTS

1. Performance: The web application should provide quick and responsive predictions, ensuring low latency for user inputs and rendering of results.

2. Accuracy: The KNN model should be trained with a high level of accuracy to provide reliable predictions to users.

3. Maintainability: The codebase and project structure should be well-organized and documented to enable easy maintenance, updates, and future enhancements.

4. Usability: The user interface should be visually appealing, with clear instructions and appropriate feedback to guide users through the prediction process.

5. Compatibility: The web application should be compatible with different web browsers and devices to reach a broader audience.

6. Reliability: The application should be stable and robust, with minimal downtime or errors during regular usage.

3.3 SOFTWARE REQUIREMENTS

Programming languages: Python

Data Analysis and Visualization libraries: NumPy, Pandas, Matplotlib

Modelling libraries: scikit-learn, pickle

HTTP Server: Gunicorn

Front-End: HTML, CSS, Bootstrap

Back-End: Flask

3.4 HARDWARE REQUIREMENTS

Processor: A modern processor like Intel Core i5 or AMD Ryzen with at least two cores is recommended. A faster processor will provide better performance during data preprocessing and model training.

RAM: At least 8 GB of RAM is recommended to handle data processing, model training, and web application execution efficiently. More RAM might be needed for larger datasets or more complex machine learning models.

Internet Connectivity: A stable internet connection is necessary for accessing online resources, installing packages, and deploying the web application.

CHAPTER 4

SYSTEM DESIGN OR METHODOLOGY

4.1 SYSTEM ARCHITECTURE

1. Problem Statement: Predicting the price of a mobile phone based on its specifications. The user should specify the input features (e.g., processor speed, RAM, camera quality) and the target variable (price) is predicted.

2. Data Collection and Preprocessing: The data is collected from Kaggle that has the following columns.

1	battery_pc	blue	clock_spe	dual_sim	fc	five_g	int_memo	m_dep	mobile_wt	n_cores
2	842	0	2.2	0	1	0	7	0.6	188	2
3	1021	1	0.5	1	0	1	53	0.7	136	3
4	563	1	0.5	1	2	1	41	0.9	145	5
5	615	1	2.5	0	0	0	10	0.8	131	6
6	1821	1	1.2	0	13	1	44	0.6	141	2
7	1859	0	0.5	1	3	0	22	0.7	164	1
8	1821	0	1.7	0	4	1	10	0.8	139	8
9	1954	0	0.5	1	0	0	24	0.8	187	4
10	1445	1	0.5	0	0	0	53	0.7	174	7
11	509	1	0.6	1	2	1	9	0.1	93	5
12	769	1	2.9	1	0	0	9	0.1	182	5
13	1520	1	2.2	0	5	1	33	0.5	177	8
14	1815	0	2.8	0	2	0	33	0.6	159	4

Figure 4.1: Part-1 of Dataset

pc	px_height	px_width	ram	sc_h	sc_w	talk_time	four_g	touch_scr	wifi	price_range
2	20	756	2549	9	7	19	0	0	1	1
6	905	1988	2631	17	3	7	1	1	0	2
6	1263	1716	2603	11	2	9	1	1	0	2
9	1216	1786	2769	16	8	11	1	0	0	2
14	1208	1212	1411	8	2	15	1	1	0	1
7	1004	1654	1067	17	1	10	1	0	0	1
10	381	1018	3220	13	8	18	1	0	1	3
0	512	1149	700	16	3	5	1	1	1	0
14	386	836	1099	17	1	20	1	0	0	0
15	1137	1224	513	19	10	12	1	0	0	0
1	248	874	3946	5	2	7	0	0	0	3
18	151	1005	3826	14	9	13	1	1	1	3
17	607	748	1482	18	0	2	1	0	0	1
11	344	1440	2680	7	1	4	1	0	1	2

Figure 4.2: Part-2 of Dataset

The description of the dataset is given as follows:

id : ID

battery_power : Total energy a battery can store in one time measured in mAh

blue : Has bluetooth or not

clock_speed : speed at which microprocessor executes instructions

dual_sim : Has dual sim support or not

fc : Front Camera mega pixels
four_g : Has 4G or not
int_memory : Internal Memory in Gigabytes
m_dep : Mobile Depth in cm
mobile_wt : Weight of mobile phone
n_cores : Number of cores of processor
pc : Primary Camera mega pixels
px_height : Pixel Resolution Height
px_width : Pixel Resolution Width
ram : Random Access Memory in Megabytes
sc_h : Screen Height of mobile in cm
sc_w : Screen Width of mobile in cm
talk_time : longest time that a single battery charge will last when you are
three_g : Has 3G or not
touch_screen : Has touch screen or not
wifi : Has wifi or not

The data preprocessing is done by handling missing values, outliers, and inconsistencies in the dataset. Then, we impute missing data or remove outliers if necessary and normalize or scale numerical features to bring them to a similar range, avoiding biases during model training. Here, the categorical variables are encoded into numerical representations.

3. Machine Learning Model Selection: The four machine learning models used in the project are: k-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, and Random Forest. These models are chosen for the task of classification of mobiles into 4 different price ranges. The 4 machine learning models are trained on the given dataset and the scores are calculated.

The obtained accuracy scores are as follows:

- KNN : 92.1
- Logistic Regression : 74.6
- Decision Tree: 81.6
- Random Forest Classifier: 87.5

4. Flask Web Application Architecture: The View is responsible for presenting the data to the user and handling user interactions. In this project, the View is implemented using the Flask web application:

a. User Interface Module: This module handles the user interface design and is responsible for rendering HTML templates that display the input form where users

can enter mobile phone specifications. It also displays the predicted price as the output.

b. Controller: The Controller acts as an intermediary between the Model and the View. It processes user input, invokes appropriate Model functions, and updates the View accordingly. In this project, the Controller is integrated within the Flask app

c. Prediction Module: This module handles user input from the input form in the user interface. It passes the specifications to the selected KNN model for price prediction and returns the predicted price, which is then displayed on the website.

CHAPTER 5

IMPLEMENTATION

The implementation of the project is done as follows:

5.1 MACHINE LEARNING IMPLEMENTATION

The features (mobile phone specifications) are stored in a variable *X* and the corresponding target variable (mobile price ranges) stored in a variable *y*. After, we split the dataset into a training set and a testing set using train-test split. Next, we import the *KNeighborsClassifier* from *scikit-learn* and create an instance of the KNN model. Then, we train the KNN model using the training data. Now, we evaluate the performance of the KNN model using the testing data. With the trained KNN model, we can now make predictions for new mobile phone specifications for any given set of inputs.

The selected machine learning model can be used through the pickle implementation for serialization and deserialization. Pickle is a Python module that allows you to save the trained machine learning model to a file and later load it back into memory when needed. This way, you can avoid retraining the model every time the web application is restarted, saving computational resources and reducing response time.

The implementation of pickle to use model in Flask App:

1. Train and Serialize the Model
2. Load the Model in Flask App
3. Use the Model for Predictions

5.2 FRONT END DEVELOPMENT

The frontend of the customer revenue prediction website is developed using HTML, CSS, and Bootstrap to create a visually appealing and responsive user interface. The HTML markup defines the structure of the web pages, while CSS styles the elements, ensuring consistent and attractive layouts. Bootstrap, a popular front-end framework, is utilized to streamline the development process and provide pre-designed components for responsive design, such as navigation bars, buttons, and modals..User-friendly forms are designed for inputting data, and real-time updates are implemented to display the latest customer revenue predictions.

5.3 BACKEND DEVELOPMENT

The backend of the mobile price prediction website is developed using Flask, a lightweight Python web framework, to handle server-side logic and data processing. Flask facilitates the integration of the frontend with the predictive model developed for mobile price prediction. When users input relevant data through the frontend, Flask receives the data, processes it, and passes it to the trained machine learning model for prediction. The backend then receives the predictions from the model and sends them back to the frontend for visualization and display. Flask also handles error handling and reporting, providing informative feedback to users in case of any issues during data processing or predictions. Additionally, the backend is responsible for managing the continuous update of the model with new data to maintain real-time predictions and insights on the frontend.

CHAPTER 6

TESTING AND RESULTS

Four Machine Learning Algorithms are evaluated based on their scores and the one with the highest score is chosen.

The obtained scores are depicted in the following table:

<i>Algorithm</i>	<i>Score</i>
KNN	92.1
Logistic Regression	74.6
Decision Tree	81.6
Random Forest	87.5

Table 6.1: Algorithms vs scores

The homepage of the website where users give the input is as follows:

Figure 6.1: The homepage of the website

Upon predicting, we are redirected to the predict page where the result is shown as follows:

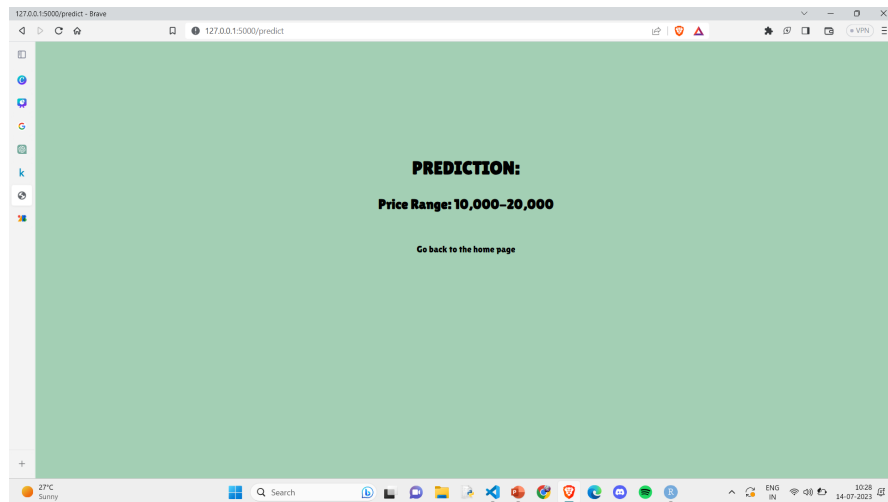


Figure 6.2: The predict page of the website

CHAPTER 7

FUTURE SCOPE

The "Mobile Price Prediction Using Machine Learning" project has several potential future scopes for further improvement and expansion:

1. Model Ensembles: Experiment with ensemble methods like stacking or blending, where multiple machine learning models, including KNN, can be combined to create a more robust and accurate prediction model. Ensembles often lead to better performance by leveraging the strengths of different algorithms.

2. Feature Engineering: Explore additional relevant features that might impact mobile prices. Feature engineering techniques like creating new features based on domain knowledge or combining existing features can enhance the model's predictive power.

3. Hyperparameter Tuning: Optimize hyperparameters of the KNN model to find the best combination for the given dataset. Techniques like grid search or random search can be employed to identify the optimal values for parameters like the number of neighbors (k) and distance metrics.

4. Advanced Machine Learning Models: Implement and compare the performance of more sophisticated machine learning algorithms, such as Support Vector Machines (SVM), Gradient Boosting Machines (GBM), or Neural Networks. These models might capture more complex patterns in the data and potentially improve accuracy.

5. Mobile Price Segmentation: Instead of predicting the price as a continuous value, consider converting the problem into a classification task to predict price segments (e.g., budget, mid-range, premium) based on the mobile specifications.

6. User Reviews and Sentiment Analysis: Incorporate user reviews and sentiment analysis to gain insights into how specific mobile phone features influence price perception and overall user satisfaction.

7. Real-Time Data Updates: Develop a mechanism to automatically update the machine learning model with new mobile phone data regularly. This ensures that the model stays relevant and accurate as new devices are released in the market.

8. Deployment on Mobile Platforms: Convert the web application into a mobile application to make mobile price predictions accessible to a broader user base on smartphones and tablets.

9. Price Forecasting: Extend the project to include price forecasting capabilities, predicting future price trends based on historical data and market trends.

10. User Personalization: Implement a user preference system that takes user preferences into account to tailor predictions according to individual tastes and requirements.

By exploring these future scopes and enhancements, the "Mobile Price Prediction Using Machine Learning" project can continue to evolve, providing more accurate and valuable insights for users interested in buying or selling mobile phones.

CHAPTER 8

CONCLUSION

In conclusion, the "Mobile Price Prediction Using Machine Learning" project successfully demonstrates the application of machine learning techniques to predict mobile phone prices based on their specifications. Through the implementation of the k-Nearest Neighbors (KNN) model and the development of a user-friendly Flask web application, the project provides a practical and accessible tool for users seeking to estimate mobile phone prices in real-time.

The project's architecture follows a well-organized structure, with clear separation of concerns between different modules, including data preprocessing, machine learning, user interface, and prediction. This modular design enables easy maintainability and future scalability.

The project's achievements include accurate price predictions and a user-friendly interface that allows users to input mobile specifications and receive instant price estimates. Additionally, the use of evaluation metrics ensures the model's performance is measurable, and ongoing improvements and future enhancements can be based on rigorous testing and analysis.

As the project evolves, the future scope encompasses further exploration of advanced machine learning techniques, feature engineering, model ensembles, and real-time data updates to ensure the model's relevance and robustness in a dynamic mobile market. The project's impact extends beyond price prediction to potential applications in mobile price segmentation, user reviews analysis, and mobile price forecasting.

In summary, the "Mobile Price Prediction Using Machine Learning" project exemplifies the fusion of data science and web development to create an intuitive tool with practical value for users interested in mobile phone pricing. With continued dedication to improvement and adaptation to emerging technologies, the project holds the potential to become an indispensable resource for users in the ever-evolving mobile industry.

APPENDIX A

CODE ATTACHMENTS

A.1 Machine Learning Model (KNN)

```
1 from os import pipe
2 import pandas as pd
3 import numpy as np
4
5
6 dataset=pd.read_csv('train.csv')
7
8
9 X=dataset.drop('price_range',axis=1)
10 y=dataset['price_range']
11
12 from sklearn.model_selection import train_test_split
13
14 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size
    =0.33, random_state=101)
15
16 from sklearn.neighbors import KNeighborsClassifier
17 knn = KNeighborsClassifier(n_neighbors=10)
18 knn.fit(X_train, y_train)
19
20 knn.score(X_test, y_test)
21
22 error_rate = []
23 for i in range(1,20):
24
25     knn = KNeighborsClassifier(n_neighbors=i)
26     knn.fit(X_train, y_train)
27     pred_i = knn.predict(X_test)
28     error_rate.append(np.mean(pred_i != y_test))
29
30
31 arr=np.array
    ([[1600,1,12,1,12,1,512,1,100,4,15,1566,1453,512,8,9,30,1,1,1]])
32 print(knn.predict(arr))
33
34 import pickle
35
36 pickle.dump(knn, open('knn.pkl', 'wb'))
```

A.2 Frontend (HTML and CSS)

A.2.1 homepage.html

```

1 <html>
2   <head>
3     <meta charset="utf-8">
4     <meta name="viewport" content="width=device-width, initial-scale
      =1">
5     <title>Mobile Price Predictor</title>
6     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/
      dist/css/bootstrap.min.css" rel="stylesheet"
7       integrity="sha384-GLhlTQ8iRABdZL16O3oVMWSktQOp6b7In1Zl3/
      Jr59b6EGGoI1aFkw7cmDA6j6gD" crossorigin="anonymous">
8   <style>
9     body {
10       background-color: #222;
11       color: #fff;
12     }
13
14     .form-group {
15       margin-bottom: 15px;
16     }
17
18     label {
19       display: inline-block;
20       margin-bottom: 5px;
21       color: green;
22     }
23
24     h1 {
25       color: green;
26     }
27   </style>
28 </head>
29
30 <body>
31   <div class="container container-sm">
32     <div class="row">
33       <div class="card mt-50" style="width: 100%; height: 100%">
34         <div class="card-header">
35           <div class="col-12" style="text-align: center;">
36             <h1>Mobile Price Predictor</h1>
37           </div>
38         </div>
39
40         <form method="POST" action="{{ url_for('home') }}">
41           <div class="row">
42             <div class="col-md-6">
43               <div class="form-group">
44                 <label for="battery_power">Enter the battery power
45                   in mAh</label>
46                 <input type="number" class="form-control" name="
47                   battery_power" id="battery_power" placeholder="
48                   Enter value" required>
49               </div>

```

```

50         <input class="form-check-input" type="radio" name
           ="blue" id="blue_yes" value="1" required>
51         <label class="form-check-label" for="blue_yes">
           Yes</label>
52     </div>
53     <div class="form-check">
54         <input class="form-check-input" type="radio" name
           ="blue" id="blue_no" value="0">
55         <label class="form-check-label" for="blue_no">No<
           /label>
56     </div>
57 </div>
58 <div class="form-group">
59     <label for="clock_speed">Enter the clock speed(the
           speed at which microprocessor executes
           instructions)</label>
60     <input type="number" class="form-control" name="
           clock_speed" id="clock_speed" placeholder="Enter
           _value" required>
61 </div>
62 <div class="form-group">
63     <label for="dual_sim">Does it support dual sim?</
           label>
64     <div class="form-check">
65         <input class="form-check-input" type="radio" name
           ="dual_sim" id="dual_sim_yes" value="1"
           required>
66         <label class="form-check-label" for="dual_sim_yes
           ">Yes</label>
67     </div>
68     <div class="form-check">
69         <input class="form-check-input" type="radio" name
           ="dual_sim" id="dual_sim_no" value="0">
70         <label class="form-check-label" for="dual_sim_no"
           >No</label>
71     </div>
72 </div>
73 <div class="form-group">
74     <label for="fc">Enter the front camera mega pixels:
           </label>
75     <input type="number" class="form-control" name="fc"
           id="fc" placeholder="Enter _value" required>
76 </div>
77 </div>
78
79 <div class="col-md-6">
80     <div class="form-group">
81         <label for="four_g">Does it support 5G?</label>
82         <div class="form-check">
83             <input class="form-check-input" type="radio" name
               ="four_g" id="four_g_yes" value="1" required>
84             <label class="form-check-label" for="four_g_yes">
               Yes</label>
85         </div>
86         <div class="form-check">
87             <input class="form-check-input" type="radio" name

```

```

88         ="four_g" id="four_g_no" value="0">
        <label class="form-check-label" for="four_g_no">
            No</label>
89    </div>
90 </div>
91 <div class="form-group">
92     <label for="int_memory">Enter the internal memory
        in gigabytes:</label>
93     <input type="number" class="form-control" name="
        int_memory" id="int_memory" placeholder="Enter _
        value" required>
94 </div>
95 <div class="form-group">
96     <label for="m_dep">Enter the mobile depth in cm:</
        label>
97     <input type="number" class="form-control" name="
        m_dep" id="m_dep" placeholder="Enter _value"
        required>
98 </div>
99 <div class="form-group">
100    <label for="mobile_wt">Enter the mobile weight in
        gms:</label>
101    <input type="number" class="form-control" name="
        mobile_wt" id="mobile_wt" placeholder="Enter _
        value" required>
102 </div>
103 <div class="form-group">
104    <label for="n_cores">Enter the number of cores in
        the mobile:</label>
105    <input type="number" class="form-control" name="
        n_cores" id="n_cores" placeholder="Enter _value"
        required>
106 </div>
107 </div>
108 </div>
109
110 <div class="row">
111     <div class="col-md-6">
112         <div class="form-group">
113             <label for="pc">Enter the primary camera mega
                pixels:</label>
114             <input type="number" class="form-control" name="pc"
                id="pc" placeholder="Enter _value" required>
115         </div>
116         <div class="form-group">
117             <label for="px_height">Enter the pixel resolution
                height:</label>
118             <input type="number" class="form-control" name="
                px_height" id="px_height" placeholder="Enter _
                value" required>
119         </div>
120         <div class="form-group">
121             <label for="px_width">Enter the pixel resolution
                width:</label>
122             <input type="number" class="form-control" name="
                px_width" id="px_width" placeholder="Enter _value

```



```

123         " required>
124     </div>
125     <div class="form-group">
126         <label for="ram">Enter the RAM size in MB</label>
127         <input type="number" class="form-control" name="ram"
128             " id="ram" placeholder="Enter_value" required>
129     </div>
130     <div class="form-group">
131         <label for="sc_h">Enter the Screen Height of mobile
132             in cm</label>
133         <input type="number" class="form-control" name="
134             sc_h" id="sc_h" placeholder="Enter_value"
135             required>
136     </div>
137 </div>
138 <div class="col-md-6">
139     <div class="form-group">
140         <label for="sc_w">Enter the Screen Width of mobile
141             in cm</label>
142         <input type="number" class="form-control" name="
143             sc_w" id="sc_w" placeholder="Enter_value"
144             required>
145     </div>
146     <div class="form-group">
147         <label for="talk_time">Enter the longest time that
148             a single battery charge will last:</label>
149         <input type="number" class="form-control" name="
150             talk_time" id="talk_time" placeholder="Enter_
151             value" required>
152     </div>
153     <div class="form-group">
154         <label for="three_g">Does it support 4G?</label>
155         <div class="form-check">
156             <input class="form-check-input" type="radio" name
157                 ="three_g" id="three_g_yes" value="1" required
158                 >
159             <label class="form-check-label" for="three_g_yes"
160                 >Yes</label>
161         </div>
162         <div class="form-check">
163             <input class="form-check-input" type="radio" name
164                 ="three_g" id="three_g_no" value="0">
165             <label class="form-check-label" for="three_g_no">
166                 No</label>
167         </div>
168     </div>
169     <div class="form-group">
170         <label for="touch_screen">Does it have a Touch
171             Screen?</label>
172         <div class="form-check">
173             <input class="form-check-input" type="radio" name
174                 ="touch_screen" id="touch_screen_yes" value="1
175                 " required>
176             <label class="form-check-label" for="
177                 touch_screen_yes">Yes</label>

```

```

159         </div>
160         <div class="form-check">
161             <input class="form-check-input" type="radio" name
162                 ="touch_screen" id="touch_screen_no" value="0"
163                 >
164             <label class="form-check-label" for="
165                 touch_screen_no">No</label>
166         </div>
167     </div>
168     <div class="form-group">
169         <label for="wifi">Does it support Wi-Fi?</label>
170         <div class="form-check form-check-inline">
171             <input class="form-check-input" type="radio"
172                 name="wifi" id="wifi_yes" value="1" required
173                 >
174             <label class="form-check-label" for="wifi_yes">
175                 Yes</label>
176         </div>
177         <div class="form-check form-check-inline">
178             <input class="form-check-input" type="radio"
179                 name="wifi" id="wifi_no" value="0">
180             <label class="form-check-label" for="wifi_no">
181                 No</label>
182         </div>
183     </div>
184     </div>
185     </div>
186     <div class="text-center">
187         <button type="submit" class="btn btn-primary">Predict!<
188             /button>
189     </div>
190 </form>
191 </div>
192 </div>
193 </div>
194 </body>
195 </html>

```

A.2.2 predict.html

```

1 <html>
2 <head>
3     <link rel="preconnect" href="https://fonts.googleapis.com">
4     <link rel="preconnect" href="https://fonts.gstatic.com"
5         crossorigin>
6     <link href="https://fonts.googleapis.com/css2?family=Lilita+One&
7         display=swap" rel="stylesheet">
8     <style>
9         body {
10             background-color: #a3cfb4;
11             /* background-image: url('path/to/background-image.jpg');
12                 */
13             background-size: cover;

```

```

11         background-position: center;
12         color: black;
13         font-family: 'Lilita One', cursive;
14     }
15
16     .container {
17         text-align: center;
18         margin-top: 200px;
19         color: black;
20     }
21
22     h1 {
23         font-size: 32px;
24         margin-bottom: 30px;
25     }
26
27     .price-range {
28         font-size: 24px;
29         margin-bottom: 20px;
30     }
31
32     a {
33         color: black;
34         text-decoration: none;
35         font-size: 16px;
36     }
37
38     a:hover {
39         text-decoration: underline;
40     }
41 </style>
42 </head>
43
44 <body>
45     <div class="container">
46         <h1>PREDICTION:</h1>
47         {% if data == 0 %}
48             <h2 class="price-range">Price Range: 10,000-20,000</h2>
49         {% elif data == 1 %}
50             <h2 class="price-range">Price Range: 20,000-30,000</h2>
51         {% elif data == 2 %}
52             <h2 class="price-range">Price Range: 30,000-40,000</h2>
53         {% elif data == 3 %}
54             <h2 class="price-range">Price Range: 40,000-50,000</h2>
55         {% else %}
56             <h2>Invalid Data</h2>
57         {% endif %}
58         <br><br>
59         <a href="/">Go back to the home page</a>
60     </div>
61 </body>
62
63 </html>

```

A.3 Backend (Flask App)

A.3.1 app.py

```
1 from flask import Flask, render_template, request
2 import pickle
3 import numpy as np
4 import warnings
5 warnings.filterwarnings("ignore")
6
7 model = pickle.load(open('knn.pkl', 'rb'))
8
9 app = Flask(__name__)
10
11
12
13 @app.route('/')
14 def man():
15     return render_template('home.html')
16
17
18 @app.route('/predict', methods=['POST'])
19 def home():
20     battery_power = request.form['battery_power']
21     blue = request.form['blue']
22     clock_speed = request.form['clock_speed']
23     dual_sim = request.form['dual_sim']
24     fc = request.form['fc']
25     four_g = request.form['four_g']
26     int_memory = request.form['int_memory']
27     m_dep = request.form['m_dep']
28     mobile_wt = request.form['mobile_wt']
29     n_cores = request.form['n_cores']
30     pc = request.form['pc']
31     px_height = request.form['px_height']
32     px_width = request.form['px_width']
33     ram = request.form['ram']
34     sc_h = request.form['sc_h']
35     sc_w = request.form['sc_w']
36     talk_time = request.form['talk_time']
37     three_g = request.form['three_g']
38     touch_screen = request.form['touch_screen']
39     wifi = request.form['wifi']
40     arr = np.array([[battery_power, blue, clock_speed, dual_sim, fc,
41                     four_g, int_memory, m_dep, mobile_wt, n_cores, pc, px_height,
42                     px_width, ram, sc_h, sc_w, talk_time, three_g, touch_screen, wifi]])
41     arr=arr.astype(float)
42     pred = model.predict(arr)
43     return render_template('after.html', data=pred)
44
45
46 if __name__ == "__main__":
47     app.run(debug=True)
```

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